

FINITE GROUPS THAT ARE THE PRODUCT OF EVERY PAIR OF NON-CONJUGATE MAXIMAL SUBGROUPS ARE SOLUBLE

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ABSTRACT. We prove that every finite group that is the product of every pair of its non-conjugate maximal subgroups is soluble, answering Problem 10.34 of the Kourovka Notebook (V. S. Monakhov, 1986) in the negative. Tikhonenko and Tyutyanov settled the almost simple case in 2010. What remains is the possibility of a minimal counterexample whose unique minimal normal subgroup is S^k , with S non-abelian simple and $k \geq 2$. Because k is unbounded, no bound depending on a fixed ratio of subgroup orders can handle all such groups. We rule out this possibility with a divisibility criterion: a single pair of automorphism-stable conjugacy classes of subgroups of S , satisfying a valuation inequality, excludes the configuration for every $k \geq 2$ and every admissible embedding. Such pairs are constructed uniformly for every infinite family of finite simple groups, with Zsigmondy primes as the arithmetic obstruction. The sporadic groups and the finitely many exceptional cases are handled by independently checkable GAP certificates.

1. INTRODUCTION

Problem 10.34 of the Kourovka Notebook, posed by V. S. Monakhov in 1986 [18], asks:

Does there exist a finite non-soluble group that coincides with the product of any two of its non-conjugate maximal subgroups?

Many soluble groups have this property: it holds vacuously whenever all maximal subgroups are conjugate, and $S_4 = AB$ for every pair of non-conjugate maximal subgroups A, B . The content of the problem is whether the property forces solubility. We prove that it does.

Theorem 1.1 (Main Theorem). *Let G be a finite group such that $G = AB$ for every pair of maximal subgroups $A, B \leq G$ that are not conjugate in G . Then G is soluble.*

Throughout, we say G has *property P* if $G = AB$ for every pair of non-conjugate maximal subgroups $A, B \leq G$.

The almost simple case of Theorem 1.1 was proved by Tikhonenko and Tyutyanov [32]. We use their theorem as the essential $k = 1$ branch of the main proof. The new content is the case $k \geq 2$. Tikhonenko and Tyutyanov also record that Zenkov had announced a negative answer to the full problem in a 1997 conference abstract [36], while stating that no proof of that announcement was known to them. Searches completed on 2 August 2026 did not identify a published proof, and Problem 10.34 remains listed among the unsolved problems in the July 2026 revision of the Kourovka Notebook [18]. We therefore claim a proof of the theorem, not priority for the conclusion. For comparison, Theorem 4.3 shows that the present criterion also recovers the almost simple conclusion; it is not used in the proof of Theorem 1.1.

The maximal factorizations of finite simple groups and their automorphism groups were classified by Liebeck, Praeger and Saxl [21]. Their tables classify the factorizations that *do* occur.

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Our route needs only to exhibit, for each coordinate closure, one pair of classes whose associated maximal subgroups cannot factorize G ; the valuation obstruction avoids importing the full maximal-factorization tables.

Strategy. A minimal counterexample G has a unique minimal normal subgroup $N = S^k$ with S non-abelian simple, and embeds in $\text{Aut}(S) \wr S_k$ (§2). For an automorphism-stable conjugacy class $[V]$ of suitable subgroups $V < S$, the normalizer $B_V = N_G(V^k)$ is a maximal subgroup of G of order exactly $|G/N| \cdot |V|^k$ (§3). Property P forces $G = B_V B_W$ for distinct such classes, and the resulting integrality condition $|B_V \cap B_W| \in \mathbb{Z}$ produces, for a well-chosen prime p , the divisibility $p^{\eta k} \mid |G/N|$ with η independent of k , which is incompatible with $|G/N|$ dividing $x^k k!$, where $x = |X/\text{Inn}(S)|$ for the relevant $X \leq \text{Aut}(S)$ (§4). The conclusion (Theorem 4.1) is that a *single* pair of stable classes in S , satisfying a valuation inequality, rules out the corresponding minimal-normal-subgroup configuration for every $k \geq 2$ and every admissible embedding.

The $k \geq 2$ branch thus reduces to exhibiting such a pair for every finite simple group S . By the classification of finite simple groups [12, Ch. 1, Table I, pp. 8–10] (see also [27, Thm. 1.1]), the possibilities for S comprise finitely many infinite families, the 26 sporadic groups, and the Tits group. For the infinite families the pairs are constructed uniformly in §6, with Zsigmondy primes of the relevant torus exponents as the obstruction primes. The finite exception manifest in [28] covers the Zsigmondy exceptions, small-field maximality exceptions, the exceptional outer automorphisms of A_6 , and the other isolated cases. The manifest identifies the relevant simple groups. The only ones above 1.05×10^7 are the four alternating groups $A_{11}, A_{12}, A_{13}, A_{14}$, handled by machine certificates, and the three Lie-type groups $\text{PSL}(6, 2)$, $\Omega^+(8, 2)$, and $\text{Sp}(4, 8)$, treated in the uniform proofs with substitute primes. In the interval $5 \times 10^5 \leq |S| \leq 1.05 \times 10^7$, the 38 groups $\text{PSL}(2, q)$ are handled by the uniform proof of Theorem 6.1; the remaining 13 groups, the 47 groups below 5×10^5 , and every sporadic group are handled by machine certificates (§5).

Three structural facts keep the family analysis manageable. First, when the two exhibited classes consist of *maximal* subgroups of S , all hypotheses of the machinery are automatic and only stability and the valuation inequality need checking. Second, for each twisted family used below, the automorphism structure induces no nontrivial permutation of the types of the relevant *relative* (twisted) diagram; field and diagonal automorphisms preserve those types. Thus the selected parabolic classes are stable under every automorphism. The twisted families, including the triality groups ${}^3D_4(q)$ and the Ree groups, are consequently among the easiest cases. Third, when a graph automorphism does fuse the relevant parabolic classes, the corresponding *flag* parabolics (intersections of fused maximal parabolics) become poset-maximal among stable classes, and a structural lemma (Lemma 3.6) shows they satisfy all needed hypotheses; this is the wreath-product analogue of the “novelty” phenomenon for maximal subgroups of almost simple groups.

Computer verification. The machine-certified ingredients were produced with GAP 4.16.0 [11], using its simple-group iterators, the character table library [2] and ATLAS data [8], in the pinned computational environment described in §5. The verification scripts used in the proof are deterministic and fail closed: any skipped case, soft failure, ambiguous identification, missing datum, version mismatch, or byte-level certificate difference is fatal to verification, and superseded exploratory scripts are not part of the verification suite. Every finite certificate records class positions and guard fingerprints as well as orders, the obstruction prime, and valuations. The scripts and logs are archived in a versioned release [28]; §5 specifies what is claimed from the machine and how it is certified, and also records the verification records behind the arithmetic of the family proofs of §6.

The structural arguments of §§2–4 and the universal arithmetic deductions used in §6 are formalized in Lean 4 and Rocq and checked by their kernels. Lean checks the quotient inheritance

and the minimal-order core of the reduction: uniqueness and characteristic simplicity of the minimal normal subgroup, non-solubility, soluble quotient, trivial centralizer, and faithfulness of the conjugation map $G \rightarrow \text{Aut}(N)$. It also checks the product-supplement intersection, exact order, maximality, and non-conjugacy statements of §3, under the normalized coordinate-action hypotheses of Conventions 2.4–2.5. A further development checks the subgroup-product/lower-divisibility bridge and the universal-in- k arithmetic contradiction of §4. Finally, Lean checks the universal arithmetic deductions of the exact 34-branch arithmetic manifest behind the family proofs, conditional on the published group, Levi, outer-automorphism, maximality, and Zsigmondy inputs identified by the source rows.

The direct-power step has a separate Rocq/MathComp kernel check: every finite characteristically simple non-soluble group is an internal direct product of $k > 0$ automorphic copies of one non-soluble, non-abelian simple subgroup S , with $|N| = |S|^k$. The strengthened Rocq theorem also constructs an isomorphism from N to the explicit external coordinate product of those same factors.

Lean then kernel-checks reindexing any such nonempty finite coordinate type to $\{1, \dots, k\}$, producing exactly the group equivalence $N \cong S^k$ used in the subsequent construction. From that equivalence the Lean development constructs the coordinate factors and proves that every automorphism permutes them; constructs the injective homomorphism $\text{Aut}(S^k) \rightarrow \text{Aut}(S) \wr S_k$; proves transitivity of the ambient G -action from minimal normality by an orbit-support argument; identifies the inverse image of $\text{Inn}(S)^k$ as N ; and checks the coordinate normalization and the divisibility $|G/N| \mid |X/\text{Inn}(S)|^k k!$.

The three theorem signatures, called `producer`, `reindex`, and `consumer` in the repository, are bound by a fail-closed formal interface manifest and checker. No unproved group-theoretic lemma is inserted at the Rocq–Lean interface. No single kernel checks the audited translation from MathComp’s subgroup isomorphism and external function product to Lean’s group equivalence and function product; that translation therefore remains an explicit trusted interface. The two kernels and their pinned libraries are trusted components as well. The formal coverage and interface manifests, together with the stated trust assumptions, are part of [28].

Generative AI was used in the research and preparation of this paper; a full declaration appears before the bibliography.

Proof and verification roadmap. The proof has five separately auditable layers; each row of Table 1 states where the claim is proved and how it is verified.

Every machine-backed claim in §5 is mapped in the repository README to its script and committed log file.

2. REDUCTION TO SOCLE ANALYSIS

Lemma 2.1 (Conjugation lemma). *If $G = AB$ for subgroups A, B , then $G = A^x B^y$ for all $x, y \in G$.*

Proof. Since $G = AB$, write $xy^{-1} = ab$ with $a \in A$, $b \in B$. Then, as subsets of G ,

$$A^x B^y = x^{-1} A x y^{-1} B y = x^{-1} A (ab) B y = x^{-1} (AB) y = x^{-1} G y = G. \quad \square$$

Consequently, property P depends only on the conjugacy classes of maximal subgroups: it says every pair of *distinct classes* has one (hence every) representative pair factorizing G .

Lemma 2.2. *Property P is inherited by quotients.*

Proof. Maximal subgroups of G/K are the images of the maximal subgroups of G containing K ; non-conjugate ones lift to non-conjugate ones, and factorizations project. $\square$

Claim	Where	Verification
Minimal counterexample has unique minimal normal subgroup S^k and embeds in $X \wr S_k$	Prop. 2.3, Convs. 2.4–2.5	Elementary structural proof; kernel-checked in Lean and Rocq/MathComp
Normalizers $B_V = N_G(V^k)$ of stable classes are maximal, non-conjugate, of order $t V ^k$	Lemmas 3.3–3.5, 3.6	Structural proof; kernel-checked in Lean; parabolic input from [6]
Divisibility criterion: one stable pair with $\eta > v_p(x)$ excludes S^k for every $k \geq 2$	Thm. 4.1	Structural proof; kernel-checked in Lean
Stable pairs exist for every infinite family	§6, Thms. 6.1–6.6	Structural proofs from published order formulas; the universal arithmetic is kernel-checked in Lean and independently reproduced by two implementations (§5)
Sporadic groups, Tits group, and the finite base are excluded	Props. 5.1, 5.2	GAP certificates with coverage cross-checks; 38 upper-range $\mathrm{PSL}(2, q)$ entries are covered by Thm. 6.1

TABLE 1. Proof and verification roadmap.

Proposition 2.3 (Minimal counterexample structure). *Let G have least order among the non-soluble groups with property P. Then G has a unique minimal normal subgroup $N = S^k$ with S non-abelian simple and $k \geq 1$, $C_G(N) = 1$, G/N is soluble, and G embeds in $\mathrm{Aut}(S) \wr S_k$ with N mapping onto $\mathrm{Inn}(S)^k$ and G acting transitively on the k coordinates.*

Proof. Every proper quotient of G has property P (Lemma 2.2), hence is soluble by minimality of $|G|$. The soluble radical R of G is trivial: otherwise G/R is non-soluble (as R is soluble) with property P, contradicting minimality. So the socle of G is a direct product of non-abelian minimal normal subgroups. If $N_1 \neq N_2$ were two of them, then G/N_1 is soluble while $N_2 \cong N_2 N_1 / N_1 \leq G/N_1$ is non-soluble, a contradiction. Hence there is a unique minimal normal subgroup $N = S^k$, S non-abelian simple. Now $C_G(N) \cap N = Z(N) = 1$ since N is a direct product of non-abelian simple groups; if $C_G(N)$ were nontrivial it would contain a minimal normal subgroup of G , necessarily N by uniqueness, contradicting $C_G(N) \cap N = 1$. So $C_G(N) = 1$, and the conjugation action embeds G into $\mathrm{Aut}(N) = \mathrm{Aut}(S) \wr S_k$ in the standard way; minimal normality of N forces transitivity on coordinates. Published references for the two standard structural inputs used here are [37, Lemma 2.10, pp. 173–174] for the automorphism group of a direct power and [22, p. 432] for the resulting transitive wreath realization in the unique-minimal-normal-subgroup setting. Finally, G/N is soluble because it is a proper quotient. $\square$

Convention 2.4 (Coordinate closure). For G as in Proposition 2.3 let G_1 be the stabilizer of the first coordinate and $\pi_1: G_1 \rightarrow \mathrm{Aut}(S)$ the projection. Set $X := \pi_1(G_1) \cdot \mathrm{Inn}(S)$, the *coordinate closure*; by coordinate-transitivity all coordinates give the same X up to conjugacy, and after conjugating inside $\mathrm{Aut}(S) \wr S_k$ we may and do assume $G \leq X \wr S_k$ with $\mathrm{Inn}(S) \leq X \leq \mathrm{Aut}(S)$.

The normalization is achieved as follows, computing in the exponent convention of Lemma 2.1 (actions composing left to right). Fix $g_j \in G$ carrying coordinate 1 to coordinate j , with $g_1 = 1$, and let $a_j \in \mathrm{Aut}(S)$ be its $1 \rightarrow j$ component. If $g \in G$ carries coordinate i to coordinate j through the component $c \in \mathrm{Aut}(S)$, then $g_i g g_j^{-1}$ stabilizes coordinate 1 with

component $a_i c a_j^{-1}$, which therefore lies in $\pi_1(G_1) \leq X$. So replacing G by $\delta G \delta^{-1}$, where $\delta := (a_1, \dots, a_k) \in \text{Aut}(S)^k$ (equivalently, conjugating by $(a_1^{-1}, \dots, a_k^{-1})$ in the exponent convention of Lemma 2.1), places every component of every element of G in X , and $a_1 = 1$ leaves X itself unchanged. Since property P and all criteria below are invariant under replacing G by a conjugate under $(a, \dots, a) \in \text{Aut}(S)^k$, statements may also be proved after normalizing X up to $\text{Aut}(S)$ -conjugacy.

Convention 2.5 (Coordinate action and arithmetic notation). For *coordinate* calculations we fix the following convention. Write $c_g(z) := gzg^{-1}$ (left conjugation), let S_i be the i -th coordinate factor of $N \cong S^k$, and let $\sigma_g \in S_k$ be the permutation determined by $c_g(S_j) = S_{\sigma_g(j)}$. There are component automorphisms $a_{g,i} \in \text{Aut}(S)$ such that

$$(c_g(z))_i = a_{g,i}(z_{\sigma_g^{-1}(i)}) \quad (z \in N, 1 \leq i \leq k) :$$

the value arriving at coordinate i comes from the *inverse-permuted* source coordinate $\sigma_g^{-1}(i)$. From $c_{gh} = c_g \circ c_h$ these data compose by

$$\sigma_{gh} = \sigma_g \sigma_h, \quad a_{gh,i} = a_{g,i} a_{h,\sigma_g^{-1}(i)}.$$

After the normalization of Convention 2.4, every $a_{g,i}$ lies in X . Throughout, c_g denotes left conjugation on elements, whereas $A^g = g^{-1}Ag$ denotes conjugation of subgroups; thus $A^g = c_{g^{-1}}(A)$. Write

$$x := |X/\text{Inn}(S)|, \quad t := |G/N|.$$

Then t divides $x^k k!$.

The following observation is not needed for the proof of Theorem 1.1 but explains why the pairs used later are of “supplement” type.

Remark 2.6 (Top-supplement pairs always factorize). If $M \supseteq N$ is maximal and B is maximal with $B \not\supseteq N$, then $BN = G$ (BN is a subgroup strictly containing the maximal B), and since $N \leq M$ we get $MB \supseteq NB = G$. Hence G has property P if and only if G/N has property P and the product of every pair of non-conjugate maximal subgroups not containing N is G .

3. PRODUCT SUPPLEMENTS

Fix G as in Conventions 2.4 and 2.5: $N = S^k \leq G \leq X \wr S_k$, $t = |G/N|$, $x = |X/\text{Inn}(S)|$. For $V \leq S$ write $[V]$ for its S -conjugacy class. An S -class is X -stable if it is invariant under the action of X , i.e. V^a is S -conjugate to V for every $a \in X$ (Inn-stability being automatic).

Definition 3.1. $\mathcal{P}_X(S) := \{[V] : 1 < V < S, N_S(V) = V, [V] \text{ } X\text{-stable}\}$, partially ordered by containment up to S -conjugacy ($[V] \leq [U]$ iff $V \leq U^s$ for some $s \in S$). Call V *normally saturating* if $\langle V^W \rangle = W$ for every overgroup $V \leq W \leq S$.

Remark 3.2. If V is a *maximal* subgroup of S and $[V]$ is X -stable, then $[V] \in \mathcal{P}_X(S)$ (maximal subgroups of a non-abelian simple group are self-normalizing), $[V]$ is a maximal element of the poset, and V is normally saturating (its only overgroups are V and S , and $\langle V^S \rangle = S$ by simplicity). Thus the lemmas below apply to maximal classes with no further hypotheses.

Lemma 3.3 (Existence). *Let $[V] \in \mathcal{P}_X(S)$ and set $B_V := N_G(V^k)$. Then*

- (1) $B_V \cap N = N_S(V)^k = V^k$;
- (2) $B_V N = G$;
- (3) $|B_V| = t \cdot |V|^k$.

Proof. (1) An element of $N = S^k$ normalizes V^k iff each coordinate normalizes V . (2) Frattini-type argument: let $g \in G$. In the coordinate-action notation of Convention 2.5, $(V^k)^g = c_{g^{-1}}(V^k)$, and since every coordinate of V^k is V ,

$$(V^k)^g = \prod_{i=1}^k a_{g^{-1},i}(V),$$

each factor is the image of V under a component automorphism $a_{g^{-1},i} \in X$. By X -stability there are $s_i \in S$ with $a_{g^{-1},i}(V) = V^{s_i}$. So $(V^k)^g$ is N -conjugate to V^k : there is $n \in N$ with $(V^k)^{gn} = V^k$, i.e. $gn \in B_V$, so $g \in B_V N$. (3) $|B_V| = |B_V N| \cdot |B_V \cap N|/|N| = t|V|^k$. $\square$

Lemma 3.4 (Maximality). *If in addition $[V]$ is a maximal element of $\mathcal{P}_X(S)$ and V is normally saturating, then B_V is maximal in G .*

Proof. Let $B_V \leq H \leq G$. Since $H \supseteq B_V$ and $B_V N = G$ we have $HN = G$, so H surjects onto G/N and in particular acts coordinate-transitively.

Step 1 (the coordinate components of H recover X). Let H_1, G_1 be the coordinate-1 stabilizers. For $g \in G_1$ write $g = hn$ ($h \in H, n \in N$); n fixes every coordinate, hence so does h , so $h \in H_1$ and $\pi_1(G_1) = \pi_1(H_1) \cdot \pi_1(N \cap G_1) = \pi_1(H_1) \cdot \text{Inn}(S)$. Therefore $\pi_1(H_1)$ and $\text{Inn}(S)$ together generate X , and a class of subgroups of S is X -stable iff it is stable under $\pi_1(H_1)$.

Step 2 (Goursat's lemma and saturation: $H \cap N$ is a product). Let $T := H \cap N \supseteq V^k$, let $W_i := \pi_i(T) \leq S$ be the coordinate projections and $K_i := T \cap S_i$ the coordinate kernels (S_i the i -th factor). Then $V \leq K_i$ (as $V^k \cap S_i = V$ in coordinate i), $K_i \leq W_i$, and $V \leq W_i$. Since K_i is a normal subgroup of W_i containing V , it contains $\langle V^{W_i} \rangle = W_i$ by saturation. So $K_i = W_i$ for every i and $T = \prod_i W_i$.

Step 3 (the W_i force $H = B_V$ or $H = G$). T is H -invariant. For $h \in H_1$, projecting $T^h = T$ to the first coordinate gives $W_1^{\pi_1(h)} = W_1$, so $[W_1]$ is $\pi_1(H_1)$ -stable, hence X -stable by Step 1. For any i , coordinate-transitivity of H supplies $h \in H$ carrying coordinate i to coordinate 1 through a component $c \in X$, and projecting $T^h = T$ gives $W_1 = W_i^c$; applying X -stability of $[W_1]$ to c^{-1} gives $[W_i] = [W_1]$. Write $[W]$ for this common X -stable class. If $W = S$ then $T = N$, so $H \supseteq N$ and $H = HN = G$. Otherwise $1 < V \leq W < S$. Form the normalizer tower $W \leq N_S(W) \leq N_S(N_S(W)) \leq \dots$; it is strictly increasing until it becomes self-normalizing, and it cannot reach S (the penultimate term would be a proper normal subgroup of the simple group S). Its terminal member U is proper, self-normalizing, and $[U]$ is X -stable ($N_S(W^a) = N_S(W)^a$, so stability propagates up the tower), i.e. $[U] \in \mathcal{P}_X(S)$ with $[U] \geq [W] \geq [V]$. Maximality of $[V]$ in $\mathcal{P}_X(S)$ forces $[U] = [V]$; then $|U| = |V|$ and $V \leq W \leq U$ gives $V = W = U$. Since $V \leq W_i$ by Step 2 and $[W_i] = [W]$ gives $|W_i| = |W| = |V|$, each $W_i = V$; so $T = \prod_i W_i = V^k$, hence $|H| = |HN| \cdot |H \cap N|/|N| = t|V|^k = |B_V|$ and $H = B_V$. $\square$

Lemma 3.5 (Non-conjugacy). *Distinct $[V] \neq [W]$ in $\mathcal{P}_X(S)$ give non-conjugate B_V, B_W .*

Proof. If $B_V^g = B_W$ then $(B_V \cap N)^g = B_W \cap N$ (as $N \leq G$), i.e. $(V^k)^g = W^k$. Reading off any coordinate, W is the image of V under an element of X composed with a coordinate permutation, so $[W] = [V^a] = [V]$ for some $a \in X$ by X -stability, a contradiction. $\square$

The next lemma supplies, structurally, all hypotheses of Lemma 3.4 for the non-maximal (“novelty”) classes used in §6: the standard parabolic subgroups whose maximal parabolic overgroups are fused by a graph automorphism. We use the notation of [6]. Here S is an untwisted simple group of Lie type with a split BN -pair of rank $\ell \geq 2$; all our applications are untwisted. Write $B = U \rtimes T$ for a Borel subgroup, with unipotent radical U and maximal torus T , and let Δ be the node set of the Dynkin diagram. For $J \subseteq \Delta$, let $Q_J \supseteq B$ be the standard

parabolic whose Levi subgroup L_J is generated by T and the root subgroups corresponding to J . The maximal parabolic omitting node i is $P_i = Q_{\Delta \setminus \{i\}}$.

Lemma 3.6 (Parabolic novelties). *Fix $J \subseteq \Delta$ and put $V := Q_J$. Suppose X is such that*

- (a) *the S -class $[Q_J]$ is X -stable, and*
- (b) *for every $J \subsetneq K \subsetneq \Delta$ the class $[Q_K]$ is not X -stable.*

Then $[V] \in \mathcal{P}_X(S)$, $[V]$ is a maximal element of $\mathcal{P}_X(S)$, and V is normally saturating. Consequently Lemmas 3.3–3.5 apply to V with no computational hypothesis.

Proof. Parabolic subgroups are self-normalizing [6, Thm. 8.3.3, p. 112], so $[V] \in \mathcal{P}_X(S)$ by (a).

Overgroups. Every subgroup containing V contains B , and every subgroup containing a Borel subgroup is a parabolic Q_K , $K \supseteq J$, of the same BN -pair [6, Thm. 8.3.2, p. 112]; moreover Q_K and $Q_{K'}$ are S -conjugate iff $K = K'$ [6, Thm. 8.3.3, p. 112].

Poset-maximality. If $[U] \in \mathcal{P}_X(S)$ with $V \leq U^s < S$ for some s , then $U^s = Q_K$ for some $J \subseteq K \subsetneq \Delta$; $K = J$ gives $[U] = [V]$, while $J \subsetneq K$ contradicts (b), since $[U] = [Q_K]$ would be X -stable.

Saturation. Let $V \leq W \leq S$, so $W = Q_K$ ($K \supseteq J$) or $W = S$; we must show $\langle V^W \rangle = W$. The normal closure of V in W contains that of B . Write $W = U_K \rtimes L_K$ (Levi decomposition; for $W = S$ read $U_K = 1$, $L_K = S$). Then $U_K \leq B$, and modulo U_K the image of B is a Borel subgroup $B_L = T \cdot (U \cap L_K)$ of L_K containing the full torus T . For each node $\alpha \in K$ the reflection representative n_α lies in L_K , and $U_{-\alpha} = U_\alpha^{n_\alpha}$. The subgroup $\langle B_L^{L_K} \rangle$ contains every L_K -conjugate of every subgroup of B_L ; in particular it contains T , each U_α for $\alpha \in K$, and each $U_{-\alpha} = U_\alpha^{n_\alpha}$. These generate L_K by the root-subgroup description of the Levi decomposition [6, §8.5, pp. 118–120]. So $\langle B^W \rangle \supseteq U_K \cdot L_K = W$. $\square$

4. THE DIVISIBILITY CRITERION

For a prime p , v_p denotes the p -adic valuation and $s_p(k)$ the digit sum of k in base p , so that $v_p(k!) = (k - s_p(k))/(p - 1)$.

Theorem 4.1 (Divisibility criterion). *Let S, X be as in Conventions 2.4 and 2.5, and let $[V] \neq [W]$ be maximal elements of $\mathcal{P}_X(S)$, both normally saturating. If some prime p satisfies*

$$\eta = \eta_p(V, W) := v_p(|S|) - v_p(|V|) - v_p(|W|) > v_p(x),$$

then for every $k \geq 2$, no group G with unique minimal normal subgroup S^k and coordinate closure X has property P.

Proof. By Lemmas 3.3–3.5, B_V and B_W are non-conjugate maximal subgroups of G , so property P forces $G = B_V B_W$ and

$$|B_V \cap B_W| = \frac{|B_V| |B_W|}{|G|} = t \cdot \left(\frac{|V| |W|}{|S|} \right)^k \in \mathbb{Z},$$

whence $p^{\eta k} \mid t$. But

$$v_p(t) \leq k v_p(x) + v_p(k!) = k v_p(x) + \frac{k - s_p(k)}{p - 1} < k(v_p(x) + 1) \leq \eta k,$$

a contradiction. $\square$

Remark 4.2. (i) The criterion is uniform in k , and this uniformity handles the infinite families. No fixed counting ratio can do so, because $(k!)^{1/k} \rightarrow \infty$ and therefore eventually exceeds every fixed ratio $|V||W|/|S|$. (ii) Only the *existence* of B_V, B_W is used; no classification of the maximal subgroups of G (or even of S) is needed. In particular the criterion is insensitive to

completeness of tabulated maximal-subgroup data: it needs only that the two exhibited subgroups are maximal, or are novelties covered by Lemma 3.6, and that their classes are stable. (Machine *verification* of stability for the finite base does use enumerated subgroup data; see the trust discussion in §5.)

The criterion has a $k = 1$ counterpart, which recovers the almost simple case.

Theorem 4.3 ($k = 1$: the almost simple case). *Let $S \leq G \leq \text{Aut}(S)$ with S non-abelian simple, and put $X := G$, so that $x := |G/S| = t$. Let $[V] \neq [W]$ be maximal elements of $\mathcal{P}_X(S)$, both normally saturating, and set $B_V := N_G(V)$, $B_W := N_G(W)$. Then:*

- (1) $B_V \cap S = V$, $B_V S = G$, and $|B_V| = t|V|$;
- (2) B_V and B_W are non-conjugate maximal subgroups of G ;
- (3) if $\eta = v_p(|S|) - v_p(|V|) - v_p(|W|) > v_p(x)$ for some prime p , then G does not have property P.

Proof. Supplement and order. By the Frattini argument via X -stability, as in Lemma 3.3, we have $B_V S = G$. Moreover, $B_V \cap S = N_S(V) = V$, and hence $|B_V| = t|V|$.

Maximality. The proof of Lemma 3.4 applies without the Goursat step; normal saturation enters only through that step and is not needed at $k = 1$ (the hypothesis is kept for uniformity with Theorem 4.1). In detail, let $B_V \leq H \leq G$ and $T := H \cap S$, an H -invariant subgroup containing V . If $T = S$ then $H \supseteq S$ and $H = HS = G$. Otherwise, since $B_V \leq H$ and $G = B_V S$, we have $G = HS$. As T is H -invariant, for $g = hs$ with $h \in H$ and $s \in S$ we have $T^g = (T^h)^s = T^s$; hence $[T]$ is G -stable. Since $X = G$, it is therefore X -stable, as in Step 1 of Lemma 3.4. The normalizer tower of T ends at a proper self-normalizing subgroup U . Stability propagates up the tower, so $[U] \in \mathcal{P}_X(S)$ and $[U] \geq [V]$. Poset-maximality of $[V]$ forces $[U] = [V]$, and $V \leq T \leq U$ then gives $T = V$. Thus $H \leq N_G(V) = B_V$.

Non-conjugacy. If $B_V^g = B_W$ with $g \in G$, then $V^g = (B_V \cap S)^g = B_W \cap S = W$, and X -stability of $[V]$ (here $X = G$) gives $[W] = [V]$, a contradiction.

The arithmetic. Property P forces $G = B_V B_W$, so $t|V||W|/|S| \in \mathbb{Z}$, i.e. $p^\eta \mid t = x$, contradicting $\eta > v_p(x)$. $\square$

Convention 4.4. From now on, “ S is excluded” means: for every X with $\text{Inn}(S) \leq X \leq \text{Aut}(S)$ there exist classes as in Theorem 4.1; consequently no finite group G with unique minimal normal subgroup S^k ($k \geq 2$, any admissible embedding) has property P. The almost simple branch $k = 1$ is supplied separately by Tikhonenko–Tyutyanov [32].

5. THE MACHINE BASE

All machine certificates were produced in a pinned computational environment: GAP 4.16.0 [11] with CTblLib 1.3.11 and AtlasRep 2.1.11, overlaid with the `ContainedConjugates` fix from GAP commit `b12f8342d641075d58fcb62cc00dd433d7b8e18`; unpatched GAP 4.16.0 is rejected. Both GAP global pseudorandom sources are reset at process start to the recorded seed 1034, so internal library choices are reproducible; proof scripts contain no explicit random witness search. The repository README in [28] maps the short sweep labels used below to the corresponding script and certificate filenames and maps every paper claim to its verification artifacts.

Proposition 5.1. *Every non-abelian simple group S with $|S| \leq 1.05 \times 10^7$ is excluded (in the sense of Convention 4.4).*

Proof. We combine the uniform $\text{PSL}(2, q)$ argument of Theorem 6.1, whose proof of these cases does not appeal to the present proposition, with the finite computation in GAP organized as the two searches below. The group inventory is cross-checked against GAP’s `SimpleGroupsIterator`

over two disjoint ranges, so that no group in range is skipped: the 47 groups with $|S| < 5 \times 10^5$ (log `verify_coverage.log`) and the 51 groups with $5 \times 10^5 \leq |S| \leq 1.05 \times 10^7$ (log `verify_coverage_big.log`). Of the latter 51, the 38 $\text{PSL}(2, q)$ entries are handled by Theorem 6.1 and its arithmetic verification rather than by a redundant GAP maximal-subgroup enumeration; sweeps J3 and J6 exhaust the other 13. The coverage check also confirms that the only two cases in this upper range not excluded by maximal-class pairs, $\text{Sp}(4, 4)$ at $x = 4$ and $\text{PSL}(5, 2)$ at $x = 2$, are excluded by the novelty certificates of item (2).

- (1) *Maximal-class pairs (sweeps J, J2, J4, J3, J5, J6).* For each simple group S treated computationally and each X with $\text{Inn}(S) \leq X \leq \text{Aut}(S)$, the program lists the X -stable classes of maximal subgroups of S and searches for pairs $([V], [W])$ and primes p satisfying the inequality of Theorem 4.1.

Certificate identity and coverage. To identify coordinate closures independently of GAP's internal class numbering, each closure is assigned a SHA-256 fingerprint derived from its action on the pinned maximal-subgroup classes, together with intrinsic quotient and kernel data. The checker verifies that every required closure appears exactly once and that each certificate is attached to one such closure. Each certificate records the selected classes, their orders, the obstruction prime, and the valuation gap. For all but six simple groups in range, a maximal-class pair is found for every X . For $S = A_5$, for example, the logged tuple $(|A_4|, |S_3|, 5, 1)$ excludes A_5^k for every k and every X . In addition, the alternating groups A_{11} – A_{14} , whose orders exceed the base bound, carry certificates of the same form for every X (sweep J5, log `sweepJ5_smallAn.log`); Theorem 6.2 uses these for $11 \leq n \leq 14$.

- (2) *Novelty pairs (sweeps K, K2, K3, K4).* Six simple groups, for the indicated coordinate closures, admit too few stable maximal classes because an outer automorphism fuses classes: $\text{PSL}(3, 2)$ and $\text{PSL}(2, 11)$ at $x = 2$; A_6 for $X/\text{Inn}(S) \in \{2_2, 2_3, 2^2\}$; $\text{PSL}(3, 4)$ for several X ; $\text{Sp}(4, 4)$ at $x = 4$; and $\text{PSL}(5, 2)$ at $x = 2$. Each is excluded by a pair involving non-maximal classes that are maximal in $\mathcal{P}_X(S)$. For $\text{PSL}(3, 2)$, $\text{PSL}(2, 11)$, A_6 , and $\text{PSL}(3, 4)$, normal saturation is verified computationally by sweep K2; for $\text{Sp}(4, 4)$ and $\text{PSL}(5, 2)$, it follows structurally from Lemma 3.6.

Normal-saturation check and trust boundary. For the four computational novelty cases, three independent procedures enumerate the relevant overgroups and embedding orbits and verify normal saturation; any disagreement causes verification to fail. Each selected subgroup class is tied to its fingerprint and normal-closure result, so same-order non-conjugate classes are not identified by order alone. These calculations assume the completeness of GAP's pinned subgroup-class enumerations. This assumption is cross-checked by the independent procedures and, for the sporadic groups, against the published ATLAS data. Sample certificates: $\text{PSL}(3, 2)$ at $x = 2$ is excluded by the pair $(S_3, 7:3)$ at $r = 2$ with logged valuation gap $2 > 1$. The case $\text{Sp}(4, 4)$ at $x = 4$ is excluded by the pair (Borel, subfield $\text{Sp}(4, 2)$) at $r = 5$ and 17. The case $\text{PSL}(5, 2)$ at $x = 2$ is excluded by the incident-flag parabolics $(Q_{\{2,3\}}, Q_{\{1,4\}})$ (the point–hyperplane and line–3-space flag stabilizers, in the standard parabolic notation of Lemma 3.6) at $r = 5$ and 31. The last two are instances of the uniform constructions of Theorem 6.6 below.

□

Proposition 5.2. *Every sporadic simple group is excluded, as is the Tits group ${}^2F_4(2)'$.*

Proof. Six of the 26 sporadic groups (M_{11} , M_{12} , M_{22} , M_{23} , J_1 , J_2) have order at most 1.05×10^7 and fall under Proposition 5.1. The remaining 20 (M_{24} , J_3 , J_4 , HS, McL, Co_1 , Co_2 , Co_3 , Suz, He, Ru, O'N, Fi_{22} , Fi_{23} , Fi_{24}' , HN, Ly, Th, B , M) and the Tits group are handled by a dedicated

computation (sweep M; script `sweepM_sporadic.g`, certificate `sweepM_sporadic.log`): stable maximal classes and pair searches use the ATLAS maximal-subgroup data via the GAP character table library [2, 8] in the pinned environment above. Every record gives the character-table identifier and `Maxes` list position. If $|\text{Out}(S)| = 1$, stability is automatic. If $|\text{Out}(S)| = 2$, the program accepts a class only when its order occurs exactly once in the complete pinned `Maxes` list; automorphisms preserve order, so that class is stable. No stored class fusion is claimed or used. Each of the resulting 42 selected class records is also bound, by group, list position, identifier, and order, to a published maximal-subgroup record. For the non-Monster sporadics these are Wilson’s corrected tables [35, §4, pp. 66–68], whose published-proof provenance is summarized in [3, §2, pp. 23–24]; the Tits entries are covered independently by [33, pp. 553–563] and [31]; and the Monster entries are given by Theorems 1.1–1.2 and Table 1 of [10, pp. 862–863]. Sample certificates: the Monster is excluded via positions 45, 46 by the pair (59:29, 41:40) at $r = 71$, and the Baby Monster via positions 29, 30 by the pair ($L_2(11).2$, 47:23) at $r = 31$. By Remark 4.2, completeness is logically needed here only for the unique-order stability argument when $|\text{Out}(S)| = 2$. The source map is machine-checked against the certificate rather than maintained only in prose. $\square$

Beyond the base, the family proofs of §6 carry their own numeric verification records: the valuation inequalities of Theorems 6.1, 6.2, and 6.4–6.6 were verified mechanically over large parameter ranges (1272 prime powers for Theorem 6.1; $n \leq 10^4$ for Theorem 6.2; 7892 parameter instances, ranks up to 25 and fields up to 3000, for Theorems 6.4–6.6), with zero failures and with the Zsigmondy-exception set as documented in the proofs; the Levi-exponent inequalities behind Theorem 6.3 are checked separately by the independent symbolic verifier in [28]. These finite-range runs are regression checks, not proofs of the statements quantified over all ranks and fields.

The release also contains an exact 34-branch arithmetic manifest for Theorems 6.1–6.6. A universal checker matches every branch to its cited order formula and Zsigmondy source, verifies the stated exponent bounds for the full parameter ranges, and reproduces the treatment of the exceptional cases. The positive-base Zsigmondy exception occurs in twelve branches, corresponding to seven distinct exception records; each is checked against the substitute-prime and finite certificates. A separate symbolic implementation independently reproduces the 34 branches, bounds, residue arguments, and exceptional valuations without reading the manifest. The same universal deductions are kernel-checked in Lean; the published group, Levi, outer-automorphism, maximality, and Zsigmondy results remain cited inputs rather than newly formalized classification theorems.

6. THE INFINITE FAMILIES

Throughout this section S is simple, X is any group with $\text{Inn}(S) \leq X \leq \text{Aut}(S)$, and $x = |X/\text{Inn}(S)|$. By Remark 3.2, when the two exhibited classes consist of maximal subgroups of S , only their X -stability and the valuation inequality require proof. We use Zsigmondy’s theorem [38, specialized theorem, p. 283] throughout: for integers $a \geq 2$, $e \geq 3$ with $(a, e) \neq (2, 6)$, there is a prime r with $\text{ord}_r(a) = e$ (a *primitive prime divisor* of $a^e - 1$); such r satisfies $r \equiv 1 \pmod{e}$. For $q = p^f$ we write $\Phi_e = \Phi_e(q)$ for cyclotomic values and always apply Zsigmondy to the prime p with exponent ef , so $\text{ord}_r(p) = ef$ and $r \equiv 1 \pmod{ef}$. The source’s additional positive-base exception has exponent 2, which lies outside every invocation here and therefore does not apply. An invocation-by-invocation audit of the hypotheses and exceptions is part of [28]. The $b = 1$ specialization and both exceptions are independently restated in [17, Thm. 2.2, p. 2].

The simple-group and outer-automorphism order formulas used below are those of [8, Tables 5–6, p. xvi]; for the classical families, the same formulas and their central divisors are also

displayed in [8, §§2.1–2.4, pp. x–xii]. As an independent published derivation and tabulation we use [7, §10, pp. 220–221, and summary table, p. 239]. For parabolics, the statement that the unipotent radical is extended by the group obtained from the complementary subdiagram (with node-orbits in the twisted case) is [8, §3.6, p. xv]; the Levi decomposition and its root-subgroup description are [6, §8.5, pp. 118–120]. Thus the cyclotomic bounds below are not inferred from numeric tests: they come from the displayed group-order formulas applied to the named Levi types. The formula-to-source and parameter audit, including each exceptional substitute, is archived in [28]. In the exceptional case over $\mathbb{F}_2$ below, $\Omega^+(8, 2)$ and $P\Omega^+(8, 2)$ denote the same simple group; we use the shorter former notation when naming that finite group. For classical groups, argument notation such as $\mathrm{Sp}(4, 2^f)$ denotes a named simple group, whereas subscript notation such as $\mathrm{Sp}_2(q)$ and $\mathrm{GL}_2(q)$ is used for Levi factors.

6.1. Projective special linear groups of rank one.

Theorem 6.1. $S = \mathrm{PSL}(2, q)$, $q = p^f \geq 4$, is excluded.

Proof. For $q \in \{4, 5, 7, 8, 9, 11\}$ this is machine-certified by the lower-range J/K certificates of §5. All other simple parameters are covered uniformly below: odd $q \geq 13$ by Case A and even $q = 2^f$, $f \geq 3$, by Case B. Recall $|S| = q(q^2 - 1)/d$ with $d = \gcd(2, q - 1)$, and $\mathrm{Out}(S) = C_d \times C_f$, so $x \mid df$: that is, $x \mid 2f$ for q odd and $x \mid f$ for q even.

Case A: q odd, $q \geq 13$. Let $U := N_S(T_+)$ and $V := N_S(T_-)$ be the normalizers of the split and nonsplit maximal tori. These are dihedral groups of orders $q - 1$ and $q + 1$, respectively. Both subgroups are maximal in S for $q \geq 13$ by Dickson’s theorem [9, 16]; see also [1, Tables 8.1–8.2, p. 376, and Table 8.7, p. 380]. All split maximal tori are S -conjugate, as are all nonsplit maximal tori, and every automorphism preserves each type (there is no graph automorphism in type A_1 ; diagonal and field automorphisms act algebraically), so $[U]$ and $[V]$ are X -stable for every X . Now $|U||V|/|S| = d/q$, so for the defining prime p : $\eta = v_p(|S|) - 0 - 0 = f$, while $x \mid 2f$ gives $v_p(x) \leq v_p(f) \leq \log_p f < f = \eta$. Theorem 4.1 applies.

Case B: $q = 2^f$, $f \geq 3$. Here $|S| = q(q^2 - 1)$ and $x \mid f$. Let U be a Borel subgroup (the normalizer of a Sylow 2-subgroup), of order $q(q - 1)$, and let $V := N_S(T_+)$ be dihedral of order $2(q - 1)$. Both U and V are maximal for $q \geq 8$ [9, 16]; see also [1, Tables 8.1–8.2, p. 376]. $[U]$ is the class of Sylow-2 normalizers, and $[V]$ is the unique class of split torus normalizers; both are X -stable. Now $|U||V|/|S| = 2(q - 1)/(q + 1)$ with $q + 1$ odd and coprime to $2(q - 1)$, so it suffices to find a prime $r \mid q + 1$ with $v_r(q + 1) > v_r(f)$. If $f \neq 3$, take r a primitive prime divisor of $2^{2f} - 1$: then $r \mid (2^{2f} - 1)/(2^f - 1) = q + 1$ and $r \equiv 1 \pmod{2f}$, so $r > f$ and $v_r(x) = 0 < v_r(q + 1) = \eta$. If $f = 3$, then $q = 8$ and $q + 1 = 9$; taking $r = 3$ gives $\eta = 2 > 1 \geq v_3(x)$. $\square$

6.2. Alternating groups.

Theorem 6.2. $S = A_n$, $n \geq 5$, is excluded.

Proof. For $n \leq 14$ this is machine-certified: the cases $n \leq 10$ fall under Proposition 5.1 (including $n = 6$, whose exceptional outer automorphisms fuse classes; all five X are handled there), and the cases $11 \leq n \leq 14$ are covered by the sweep-J5 certificates of §5. Assume $n \geq 15$, so $\mathrm{Aut}(A_n) = S_n$ and $x \mid 2$.

Let p be the largest prime $\leq n$. We first record the bound $p \geq n/2 + 3$. By Nagura’s theorem [26, Theorem, pp. 180–181], there is a prime in $(y, 6y/5]$ for $y \geq 25$. Thus, for $n \geq 31$, there is a prime in $(5n/6, n]$, and $5n/6 > n/2 + 3$ for $n > 18$. For $15 \leq n \leq 30$, the largest prime $p \leq n$ satisfies the bound by inspection: $p = 13, 13, 17, 17, 19, 19, 19, 19, 23, 23, 23, 23, 23, 29, 29$ for $n = 15, \dots, 30$. In particular, $n/2 < p \leq n$.

For $m < n/2$ let $M_m := (S_m \times S_{n-m}) \cap A_n$, and for n even let $I := (S_{n/2} \wr S_2) \cap A_n$. Each M_m is maximal in A_n , and I is maximal except in a finite list of cases, all with $n \leq 14$ [20, Thm. 1 and Tables I–II, pp. 366–368]. Conjugation by S_n preserves the partition shape (resp. block structure), each shape is a single A_n -class, and $\text{Aut}(A_n) = S_n$ for $n \neq 6$: all these classes are X -stable for every X .

For any two distinct classes among $\{M_m : n - p < m < n/2\} \cup \{I \text{ if } n \text{ even and } n/2 < p\}$, we have $v_p(|A_n|) = 1$, while v_p of each subgroup order is 0 (all parts of the partitions involved are $< p$), and $v_p(x) \leq v_p(2) = 0$. Thus $\eta = 1 > 0$, and Theorem 4.1 applies, provided two such classes exist. The number of admissible intransitive m is $p - \lfloor n/2 \rfloor - 1$, plus one for I when n is even; two classes exist because $p \geq n/2 + 3$. $\square$

6.3. Lie type without graph symmetry.

Theorem 6.3. *Let S be a simple group of Lie type of rank ≥ 2 over $\mathbb{F}_q$, $q = p^f$, of a type with no symmetry of its Dynkin diagram in the given characteristic:*

$$C_n = \text{PSp}(2n, q) \ (n \geq 2, \ p \text{ odd if } n = 2), \quad B_n = \Omega(2n+1, q) \ (n \geq 3, \ q \text{ odd}),$$

$$G_2(q) \ (p \neq 3), \quad F_4(q) \ (p \neq 2), \quad E_7(q), \quad E_8(q).$$

Then S is excluded.

Proof. Write h for the relevant torus exponent: $h = 2n$ for B_n/C_n , and $h = 6, 12, 18, 30$ for G_2, F_4, E_7, E_8 .

Let P_1, P_2 be maximal parabolic subgroups from two distinct classes (rank ≥ 2 guarantees these; take the ends of the diagram). They are maximal subgroups of S : any overgroup of a parabolic contains a Borel subgroup, and every subgroup containing a Borel is parabolic [6, Thm. 8.3.2, p. 112].

Stability. $\text{Aut}(S)$ is generated by inner, diagonal, field and graph automorphisms [29], [4, Def. 3.1 and Thm. 3.4, pp. 31–32]. Diagonal and field automorphisms preserve types of vertices of the building, hence each parabolic class; type-permuting automorphisms arise only from diagram symmetries, absent for the listed types in the listed characteristics (the exceptional symmetries of B_2/C_2 , F_4 in characteristic 2 and G_2 in characteristic 3 are excluded by hypothesis and treated in Theorem 6.6). So every parabolic class is X -stable for every X .

The obstruction prime. Let r be a primitive prime divisor of $p^{hf} - 1$. Then: (i) $r \mid |S|$, since the order formula of each listed type contains the factor $q^h - 1$ (equivalently $\Phi_h(q)$) [8, Table 6, p. xvi]. (ii) r divides no maximal parabolic order: $|P_i| = q^{N_i} |L_i| (q-1)^{c_i}$ with L_i the semisimple part of the Levi factor, whose order is a product of a q -power and factors $q^j - 1$ with $j \leq h_i$, h_i the largest torus exponent of L_i ; in every listed case every proper Levi has $h_i < h$ (B_n/C_n : Levi types $A_{i-1} \times C_{n-i}$ give $j \leq \max(i, 2(n-i)) < 2n$; G_2 : Levis A_1 , $j \leq 2$; F_4 : Levis $B_3, C_3, A_2 \times A_1$, $j \leq 6$; E_7 : Levis $E_6, D_6, A_6, \dots$, exponents ≤ 12 ; E_8 : Levis E_7, D_7, A_7 , exponents ≤ 18). By primitivity r divides none of these, nor q , nor $q-1$. (iii) $r \nmid x$: $r \equiv 1 \pmod{hf}$ with $h \geq 4$, so $r > hf \geq 4f \geq d_S f \geq x$, where $d_S \leq \gcd(2, q-1)^2 \leq 4$ bounds the diagonal part and there is no graph part.

Thus $\eta = v_r(|S|) \geq 1 > 0 = v_r(x)$ for the pair $([P_1], [P_2])$, and Theorem 4.1 applies, provided r exists.

Zsigmondy exceptions. $p^{hf} = 2^6$ forces $p = 2$ and $hf = 6$: either $G_2(2)$ (not simple; $G_2(2)' \cong \text{PSU}(3, 3)$ is in the machine base) or $\text{Sp}(6, 2)$ (machine base). For $h \in \{12, 18, 30\}$, $hf = 6$ is impossible; B_n has q odd. $\square$

6.4. Twisted groups of twisted rank at least two.

Theorem 6.4. *Let S be one of*

$$\begin{aligned} & \text{PSU}(n, q) \ (n \geq 4), \quad P\Omega^-(2n, q) = {}^2D_n(q) \ (n \geq 4), \\ & {}^3D_4(q), \quad {}^2E_6(q), \quad {}^2F_4(q) \ (q = 2^f, \ f \text{ odd} \geq 3). \end{aligned}$$

Then S is excluded.

Proof. Each of these groups has a twisted BN -pair of rank at least 2, and therefore has at least two classes of maximal parabolic subgroups P_1, P_2 . These subgroups are maximal in S , by the same argument as in Theorem 6.3: every finite group of Lie type in defining characteristic has a split BN -pair [15, Example 1.16 and §1.3.5, pp. 10–11], and Carter’s overgroup and self-normalizer theorems are stated for an arbitrary group with a BN -pair [6, Thms. 8.3.2–8.3.3, p. 112].

Stability. We claim every parabolic class is X -stable for every X . Every $a \in \text{Aut}(S)$ permutes the parabolic subgroups: they are exactly the overgroups of the normalizers of the Sylow p -subgroups (Borel subgroups), an automorphism-invariant description; hence a permutes the finitely many parabolic classes, preserving inclusion up to conjugacy, i.e. acts on the type set of the twisted building. Now use the automorphism description of [4, Def. 3.3 and Thm. 3.4, p. 32], which sets the separate graph factor equal to 1 for a twisted group. The inner-diagonal and Frobenius representatives centralizing the defining Steinberg endomorphism preserve each orbit of simple roots, hence each node of the relative (twisted) diagram. Inner automorphisms fix classes, while the standard outer representatives normalize the standard Borel and torus and therefore preserve the class of each relative type. Thus they map every standard parabolic Q_J to a conjugate of the same type. Hence every automorphism fixes every parabolic class. (For the rank-two group ${}^2F_4(q)$ one can also see directly that the two maximal parabolic classes cannot be interchanged: their Levi factors ${}^2B_2(q)$ and $\text{SL}_2(q)$ are non-isomorphic. Note that a diagram-asymmetry argument alone would not suffice at twisted rank two: the twisted diagrams of $\text{PSU}(4, q)$ and $\text{PSU}(5, q)$ have a symmetric underlying graph, and the untwisted example $\text{Sp}(4, 2^f)$ shows that a multiple bond by itself does not prevent a type-swapping automorphism.)

The obstruction prime. For each type, choose the exponent e and let r be a primitive prime divisor of $p^e - 1$. In each case r divides $|S|$ but neither $|P_1|$ nor $|P_2|$:

- $\text{PSU}(n, q)$, n odd: $e = 2nf$, so $r \mid \Phi_{2n}(q) \mid q^n + 1$, which divides $|\text{SU}(n, q)| = q^{n(n-1)/2} \prod_{i=2}^n (q^i - (-1)^i)$. The Levi factor of P_i ($i = 1, 2$) is a central product of $\text{GL}_i(q^2)$ and $\text{GU}(n - 2i, q)$: its p' -order is a product of factors $q^{2j} - 1$ ($j \leq 2$) and $q^j - (-1)^j$ ($j \leq n - 2$), all dividing $p^m - 1$ with $m \leq \max(4f, 2(n - 2)f) < 2nf$.
- $\text{PSU}(n, q)$, n even: $e = 2(n - 1)f$, $r \mid \Phi_{2(n-1)}(q) \mid q^{n-1} + 1$ (the $i = n - 1$ factor). Levi exponents are $\leq \max(4f, 2(n - 3)f, (n - 2)f) < 2(n - 1)f$ for $n \geq 4$.
- ${}^2D_n(q)$, $n \geq 4$: $e = 2nf$, $r \mid \Phi_{2n}(q) \mid q^n + 1$, which divides $q^{n(n-1)}(q^n + 1) \prod_{i=1}^{n-1} (q^{2i} - 1)$. The Levi factor of P_i has type $\text{GL}_i(q) \times O^-(2(n - i), q)$, and its relevant exponents are at most $\max(2f, 2(n - 1)f) < 2nf$.
- ${}^3D_4(q)$: $e = 12f$, $r \mid \Phi_{12}(q) = q^4 - q^2 + 1$, which divides $q^8 + q^4 + 1 = (q^4 + q^2 + 1)(q^4 - q^2 + 1)$. Since $|{}^3D_4(q)| = q^{12}(q^8 + q^4 + 1)(q^6 - 1)(q^2 - 1)$, we have $r \mid |S|$. The two Levi factors are $\text{SL}_2(q^3) \cdot (q - 1)$ and $\text{SL}_2(q) \cdot (q^3 - 1)$ (mod centers): p' -orders divide $(q^6 - 1)(q - 1)$ and $(q^2 - 1)(q^3 - 1)$, exponents $\leq 6f < 12f$.
- ${}^2E_6(q)$: $e = 18f$, $r \mid \Phi_{18}(q) \mid q^9 + 1$, which divides $q^{36}(q^{12} - 1)(q^9 + 1)(q^8 - 1)(q^6 - 1)(q^5 + 1)(q^2 - 1)$. The four maximal parabolic classes correspond to the four τ -orbits of nodes of E_6 (τ the order-two symmetry): $\{2\}$, $\{4\}$, $\{1, 6\}$, $\{3, 5\}$; the complementary τ -stable subdiagrams are A_5 , $A_2 \times A_1 \times A_2$, D_4 , $A_2 \times A_1 \times A_1$ with induced twists, so the Levi derived groups are $\text{SU}(6, q)$, $\text{SL}_2(q) \times \text{SL}_3(q^2)$, ${}^2D_4(q)$ -type, and $\text{SL}_3(q) \times \text{SL}_2(q^2)$, with cyclotomic exponents $\leq 10f, 6f, 8f, 4f$ respectively, all $< 18f$.

Any two classes serve as P_1, P_2 . This diagram derivation agrees with the published list $({}^2A_5(q), {}^2D_4(q), A_1(q) \times A_2(q^2), A_1(q^2) \times A_2(q))$ in [25, Lemma 6.8, p. 82].

- ${}^2F_4(q)$, f odd ≥ 3 : $e = 12f$, $r \mid \Phi_{12}(q) \mid q^6 + 1$, which divides $|{}^2F_4(q)| = q^{12}(q^6 + 1)(q^4 - 1)(q^3 + 1)(q - 1)$. The two Levi derived groups are ${}^2B_2(q) \cong \text{Sz}(q)$ and $\text{SL}_2(q)$ [34, Thm. 4.5(i)–(ii), p. 166] and [23, Main Theorem, pp. 52–53]: p' -orders divide $(q^2 + 1)(q - 1)^2$ and $(q^2 - 1)(q - 1)$, exponents $\leq 4f < 12f$. (${}^2F_4(2)$ is not simple; its derived group, the Tits group, is in the machine base.)

$r \nmid x$. In every case x divides $d_S \cdot cf$, where the diagonal order d_S divides $q + 1$ (types 2A , 2E_6) or 4 (type 2D) or is 1 (3D_4 , 2F_4), and the field part has order cf with $c \leq 3$. Since $r \equiv 1 \pmod{e}$ with $e \geq 6f$, r exceeds the field-part order. If $r \mid d_S$, then either $r \mid q + 1$, which would give $\text{ord}_r(p) \mid 2f < e$, or $r \leq 4$, contradicting $r > e \geq 6$. Hence $\eta = v_r(|S|) \geq 1 > 0 = v_r(x)$, and Theorem 4.1 applies.

Zsigmondy exceptions. r fails to exist only if $p^e = 2^6$. Here $e \geq 6$ always, and $e = 6$ occurs only for $\text{PSU}(4, q)$ with $f = 1$; then $p^6 = 2^6$ forces $S = \text{PSU}(4, 2) \cong \text{PSp}(4, 3)$ [5, Thm. 5.3, p. 61]; this group, of order 25920, is settled by the lower-range certificate $\text{Sp}43$. No other case reaches $p^e = 2^6$. $\square$

6.5. Twisted groups of twisted rank one.

Theorem 6.5. *Let S be one of*

$$\text{PSU}(3, q) \ (q \geq 3), \quad \text{Sz}(q) = {}^2B_2(q) \ (q = 2^f, \ f \text{ odd} \geq 3), \quad {}^2G_2(q) \ (q = 3^f, \ f \text{ odd} \geq 3).$$

Then S is excluded.

Proof. Each group has a single class of parabolic subgroups, namely the Borel subgroups $B = N_S(Q)$, $Q \in \text{Syl}_p(S)$; B is a *maximal* subgroup (rank-one BN -pair) and its class is X -stable for every X . We pair it with a second X -stable maximal class avoiding a Zsigmondy prime.

$\text{PSU}(3, q)$, $q \geq 8$. $|S| = q^3(q^3 + 1)(q^2 - 1)/d$, $d = \gcd(3, q + 1)$; $|B| = q^3(q^2 - 1)/d$. Let V be the stabilizer of a nonisotropic point of the unitary geometry. It has order $q(q - 1)(q + 1)^2/d$ and is maximal for $q \geq 3$, $q \neq 5$ [1, Table 8.5, p. 379] (class C_1). These stabilizers form a single class, preserved by every semilinear automorphism. Take r a primitive prime divisor of $p^{6f} - 1$. It exists because the exceptional equation $p^{6f} = 2^6$ would force $q = 2$, for which S is not simple. Then $r \mid \Phi_6(q) = q^2 - q + 1$, which divides $(q^3 + 1)/(q + 1)$, and hence $r \mid |S|$; $\text{ord}_r(p) = 6f$ rules out r dividing $q(q^2 - 1)$ or $q(q - 1)(q + 1)^2$. Also $x \mid 2df$, and $r \equiv 1 \pmod{6f}$ gives $r > 2f$, while $r \mid d \mid q + 1$ would force $\text{ord}_r(p) \mid 2f$. Thus $\eta := v_r(|S|) \geq 1 > 0 = v_r(x)$. For $q \in \{3, 4, 5, 7\}$ (in particular the maximality exception $q = 5$), S is in the machine base.

$\text{Sz}(q)$. $|S| = q^2(q^2 + 1)(q - 1)$; $|B| = q^2(q - 1)$. Let $V := D_{2(q-1)}$ be the normalizer of a cyclic subgroup of order $q - 1$. It is maximal by Suzuki's classification [30]; see also [1, Thm. 7.3.5, p. 367, and Table 8.16, p. 384]. These normalizers form a single class, stable under $\text{Aut}(S) = S \rtimes C_f$. Let r be a primitive prime divisor of $2^{4f} - 1$. Such a prime exists because $4f \geq 12$. Then $r \mid \Phi_4(q) = q^2 + 1$, so $r \mid |S|$, $r \nmid |B|$, $r \nmid |V|$; $x \mid f$, and $r \equiv 1 \pmod{4f}$ gives $r > f$. Thus $v_r(|S|) \geq 1 > 0 = v_r(x)$.

${}^2G_2(q)$. $|S| = q^3(q^3 + 1)(q - 1)$; $|B| = q^3(q - 1)$. Let $V := C_S(\iota) \cong \langle \iota \rangle \times \text{PSL}(2, q)$ for an involution ι . The involutions form a single S -class, so $[V]$ is $\text{Aut}(S)$ -stable. Moreover, $|V| = q(q^2 - 1)$, and V is maximal [19, Thm. C, pp. 33–34]. Let r be a primitive prime divisor of $3^{6f} - 1$. Such a prime exists because $6f \geq 18$. Then $r \mid \Phi_6(q) = q^2 - q + 1$, which divides $(q^3 + 1)/(q + 1)$, and hence $r \mid |S|$; $\text{ord}_r(3) = 6f$ rules out $r \mid q^3(q - 1)$ and $r \mid q(q^2 - 1)$; $x \mid f < r$. So $v_r(|S|) \geq 1 > 0 = v_r(x)$. (${}^2G_2(3)$ is not simple; ${}^2G_2(3)' \cong \text{PSL}(2, 8)$ is covered by Theorem 6.1.) $\square$

6.6. Untwisted types with graph symmetry.

Theorem 6.6. *Let S be one of*

$$\begin{aligned} & \text{PSL}(n, q) \ (n \geq 3), \quad P\Omega^+(2n, q) = D_n(q) \ (n \geq 4), \quad E_6(q), \\ & \text{Sp}(4, q) \ (q = 2^f, \ f \geq 2), \quad F_4(q) \ (q = 2^f), \quad G_2(q) \ (q = 3^f). \end{aligned}$$

Then S is excluded.

Proof. We keep the standard-parabolic notation Q_J of Lemma 3.6. The *graph part* of X is the image of X in the quotient of $\text{Out}(S)$ by its inner-diagonal and field part: a subgroup of $C_2 = \langle \delta \rangle$ for $\text{PSL}(n, q)$ ($n \geq 3$), D_n ($n \geq 5$), E_6 ; a subgroup of S_3 (triality) for D_4 ; while for $\text{Sp}(4, 2^f)$, $F_4(2^f)$, $G_2(3^f)$, $\text{Out}(S)$ is cyclic of order $2f$ generated by the exceptional graph-field automorphism ρ with $\rho^2 = \varphi_p$ [4, Def. 3.1(b), p. 31, and Thm. 3.4, p. 32], and “ X has graph part” means the image of X in $\text{Out}(S)$ is not contained in $\langle \rho^2 \rangle$. An automorphism acts on the building permuting vertex types by a diagram symmetry; automorphisms with trivial graph part preserve every type. By Convention 2.4 we may normalize the graph part up to $\text{Aut}(S)$ -conjugacy; for D_4 this lets us assume a C_2 graph part is generated by the symmetry fixing nodes 1, 2 and swapping 3, 4.

Case A: X with trivial graph part. Every parabolic class is X -stable, and the argument of Theorem 6.3 applies. The selected pairs and obstruction primes are given below; the $E_6(q)$ entry refers forward to a construction valid for every X .

- $\text{PSL}(n, q)$: pair $([P_1], [P_2])$, r a primitive prime divisor of $p^{nf} - 1$. Then $r \mid \Phi_n(q) \mid q^n - 1$, and this cyclotomic factor divides $|S| \cdot d$, where $d = \gcd(n, q - 1)$. The Levi factors, of $\text{GL}_{n-1}(q)$ -type and $\text{GL}_2 \times \text{GL}_{n-2}$ -type, have cyclotomic exponents at most $(n-1)f < nf$. Moreover, $x \mid 2df$. The congruence $r \equiv 1 \pmod{nf}$ rules out $r \mid 2f$, since $r > nf \geq 3f > 2f$, and also rules out $r \mid d$, since that would imply $r \mid q - 1$ and hence $\text{ord}_r(p) \mid f$. The Zsigmondy exceptions occur when $p^{nf} = 2^6$. If $(n, f) = (3, 2)$, then $S = \text{PSL}(3, 4)$, which is in the machine base for all ten X . If $(n, f) = (6, 1)$, then $S = \text{PSL}(6, 2)$; take instead $r = 31$ ($\text{ord}_{31}(2) = 5$) and the pair $([P_2], [P_3])$. Their Levi types $\text{GL}_2 \times \text{GL}_4$ and $\text{GL}_3 \times \text{GL}_3$ have exponents at most $4 < 5$, so $31 \nmid |P_2||P_3|$, whereas $v_{31}(|\text{PSL}(6, 2)|) = 1$ and $x \mid 2$.
- $D_n(q)$, $n \geq 4$: pair $([P_1], [P_2])$ (*singular-point and singular-line stabilizers*). Let r be a primitive prime divisor of $p^{2(n-1)f} - 1$. Then $r \mid \Phi_{2n-2}(q)$, which divides $q^{n(n-1)}(q^n - 1) \prod_{i=1}^{n-1} (q^{2i} - 1)$. The Levi factors, of D_{n-1} -type and $A_1 \times D_{n-2}$ -type, have exponents at most $\max((n-1)f, 2(n-2)f) < 2(n-1)f$. Moreover, $x \mid 24f$, accounting for diagonal, field, and graph factors of orders at most 4, f , and 6, respectively. Since $r \equiv 1 \pmod{2(n-1)f}$ and $2(n-1) \geq 6$, we have $r > 6f$ and $r \nmid 24$. The only Zsigmondy exception is $p^{2(n-1)f} = 2^6$, which gives $(n, f) = (4, 1)$ and $S = \Omega^+(8, 2)$; Case B3 handles this group for every X .
- $E_6(q)$. The pair used in Case B4 is stable for every X .
- $\text{Sp}(4, 2^f)$. Take the pair $([P_1], [P_2])$, and let r be a primitive prime divisor of $2^{4f} - 1$. Such a prime exists because $4f \geq 8$. It divides $\Phi_4(q) = q^2 + 1$. The Levi factors $\text{Sp}_2(q)$ and $\text{GL}_2(q)$ have exponents at most $2f < 4f$; here $x \mid f$ and $r > 4f$.
- $F_4(2^f)$. Take the pair $([P_1], [P_2])$, and let r be a primitive prime divisor of $2^{12f} - 1$. It divides $\Phi_{12}(q)$. All proper Levi factors, of types B_3 , C_3 , $A_2 \times A_1$, and $A_1 \times A_2$, have exponents at most $6f < 12f$, and $x \mid f < r$. This completes the $p = 2$ case excluded from Theorem 6.3.
- $G_2(3^f)$. Take the pair $([P_1], [P_2])$, and let r be a primitive prime divisor of $3^{6f} - 1$. It divides $\Phi_6(q)$. Levi factors of type A_1 have exponents at most $2f < 6f$, and $x \mid f < r$. This is the $p = 3$ case of G_2 .

Case B: X with nontrivial graph part. Novelty classes below are standard parabolics Q_J handled by Lemma 3.6: each $[Q_J]$ is X -stable because J is invariant under the relevant diagram symmetry (an automorphism with graph part γ maps $[Q_J]$ to $[Q_{\gamma(J)}]$), and its proper parabolic overgroups fall into nontrivial γ -orbits, giving hypothesis (b) of Lemma 3.6; so B_{Q_J} is maximal in G by Lemma 3.4, and pairs of distinct classes are non-conjugate by Lemma 3.5. The primes r are those of Case A unless stated; r -avoidance for Q_J follows from its own Levi order (a product of cyclotomic factors of the Levi of type J and powers of $q - 1$), as listed.

B1. $\mathrm{PSL}(3, q)$, $q \geq 5$. (For $q \leq 4$ the group is in the machine base, including all ten X for $\mathrm{PSL}(3, 4)$.) The pair consists of $[B]$ and $[N_T]$, where $N_T := N_S(T) \cong ((q - 1)^2/d) \cdot S_3$ is the normalizer of a maximal split torus and $d = \gcd(3, q - 1)$. The Borel class is a novelty: its proper overgroups P_1, P_2 are swapped by δ , which exchanges the points and lines of $PG(2, q)$, so Lemma 3.6 applies. The torus normalizer is maximal for $q \geq 5$ [24, 14], [1, Table 8.3, p. 378] (class C_2). Its conjugates form a single class preserved by every automorphism. Let r be a primitive prime divisor of $p^{3f} - 1$. Then $r \mid \Phi_3(q) = q^2 + q + 1$ and $v_r(|S|) \geq 1$. Primitivity gives $r \nmid |B| = q^3(q - 1)^2/d$ and $r \nmid q - 1$, and $r \nmid |N_T| = 6(q - 1)^2/d$ follows because $r \equiv 1 \pmod{3}$ forces $r \geq 7$, so $r \nmid 6$. Finally, $x \mid 2df$: $r > 3f$ rules out $r \mid 2f$, and $r \geq 7$ rules out $r \mid d = 3$. The Zsigmondy exception $p^{3f} = 2^6$ is $\mathrm{PSL}(3, 4)$, in the machine base.

B2. $\mathrm{PSL}(n, q)$, $n \geq 4$. For $n = 4$ the pair is $([P_2], [Q_{\{2\}}])$. Node 2 is δ -fixed, so $[P_2]$ is an X -stable maximal class (Levi $A_1 \times A_1$, exponents $\leq 2f$), while $Q_{\{2\}}$, the stabilizer of an incident point-hyperplane flag, is a novelty whose proper overgroups P_1, P_3 are swapped by δ (Lemma 3.6); its Levi (A_1 plus torus) has exponents $\leq 2f$. Let r be a primitive prime divisor of $p^{4f} - 1$. It avoids both orders because $2f < 4f$, and $v_r(|S|) \geq 1$. Here $x \mid 2df$, where $d = \gcd(4, q - 1)$; the congruence $r \equiv 1 \pmod{4f}$ excludes both $r \mid 2f$ and $r \mid d$. There is no Zsigmondy exception ($2^{4f} = 2^6$ is impossible). For $n \geq 5$ the pair consists of the two novelties $([Q_{J_1}], [Q_{J_2}])$, where $J_1 := \Delta \setminus \{1, n-1\}$ and $J_2 := \Delta \setminus \{2, n-2\}$. The subgroups Q_{J_1} and Q_{J_2} are, respectively, the stabilizers of an incident point-hyperplane flag and an incident line- $(n-2)$ -space flag. Both deleted node sets $\{1, n-1\}$ and $\{2, n-2\}$ are δ -orbits, so both $[Q_{J_i}]$ are X -stable; the proper overgroups of Q_{J_1} are P_1, P_{n-1} , swapped by δ , and those of Q_{J_2} are P_2, P_{n-2} , swapped by δ and distinct classes since $n \geq 5$. The Levi types are the blocks $(1, n-2, 1)$, i.e. GL_{n-2} plus torus, with exponents $\leq (n-2)f$, and the blocks $(2, n-4, 2)$, with exponents $\leq \max(2, n-4)f$. The prime r of Case A avoids both, and $v_r(x) = 0$ as there. The Zsigmondy exception is $(n, f) = (6, 1)$, $S = \mathrm{PSL}(6, 2)$: take $r = 31$ again; the two Levi orders involve only $\mathrm{GL}_4(2)$ - and $\mathrm{GL}_2(2)$ -factors and 2-powers, so 31 divides neither, while $v_{31}(|S|) = 1 > 0 = v_{31}(2)$.

B3. $D_n(q)$. For $n \geq 5$ the graph part is at most the C_2 swapping the two spinor nodes $n-1, n$ and fixing $1, \dots, n-2$: the Case A pair $([P_1], [P_2])$ is X -stable for every X and Case A's argument goes through unchanged (its x -bound already allowed the graph factor). For $n = 4$, if the graph part is trivial or the normalized C_2 (fixing nodes 1, 2), the pair $([P_1], [P_2])$ is X -stable and Case A applies. If the graph part contains a triality, take the pair $([P_2], [Q_{\{2\}}])$: node 2 is fixed by all of S_3 , so $[P_2]$ is an X -stable maximal class (Levi $A_1 \times A_1 \times A_1$, exponents $\leq 2f$), and $Q_{\{2\}}$ (Levi A_1 plus torus, exponents $\leq 2f$) has proper overgroups $Q_{\{1,2\}}, Q_{\{2,3\}}, Q_{\{2,4\}}$ and P_4, P_3, P_1 , which fall into the two regular (triality)-orbits induced by the permutation of the end nodes $\{1, 3, 4\}$; none is X -stable (Lemma 3.6). Let r be a primitive prime divisor of $p^{6f} - 1$. It avoids both classes, $v_r(|S|) \geq 1$, and $r \nmid x$ as in Case A. *Zsigmondy exception* ($f = 1$, $p = 2$): $S = \Omega^+(8, 2)$. Here $x \mid 24$; take $r = 5 = \Phi_4(2)$, so $v_5(|S|) = v_5(2^{12}(2^6 - 1)(2^4 - 1)^2(2^2 - 1)) = 2$ and $v_5(x) \leq v_5(24) = 0$. In the triality case the pair $([P_2], [Q_{\{2\}}])$ still serves: both Levi orders are $\{2, 3\}$ -numbers, so $\eta = 2 - 0 - 0 = 2 > 0$. In the trivial/ C_2 case the pair $([P_1], [P_2])$ gives $v_5(|P_1|) = v_5(|\mathrm{SL}_4(2)|) = 1$ (Levi $D_3 \cong A_3$) and $v_5(|P_2|) = 0$, so $\eta = 2 - 1 - 0 = 1 > 0$.

B4. $E_6(q)$, all X . The diagram involution fixes nodes 2 and 4 (Bourbaki numbering: δ swaps $1 \leftrightarrow 6$, $3 \leftrightarrow 5$). The pair is $([P_2], [P_4])$: two X -stable classes of *maximal* parabolics for every X . Let r be a primitive prime divisor of $p^{12f} - 1$. Then $r \mid \Phi_{12}(q)$, which divides $|S| \cdot d$. For these classes, the Levi of P_2 has type A_5 and exponents at most $6f$; that of P_4 has type $A_2 \times A_1 \times A_2$ and exponents at most $3f$. Both bounds are less than $12f$. Finally, $x \mid 2df \leq 6f < r$, where $d = \gcd(3, q - 1)$. There is no Zsigmondy exception. This simultaneously covers Case A for E_6 .

B5. $\mathrm{Sp}(4, q)$, $q = 2^f$, $f \geq 2$, X with graph part. (For $f = 1$ the group $\mathrm{Sp}(4, 2)$ is not simple; its derived group A_6 is in the machine base.) The pair consists of $[B]$ and $[V]$. The Borel class is a novelty: its proper overgroups P_1, P_2 , the point and totally-isotropic-line stabilizers, are swapped by ρ , so Lemma 3.6 applies. For the second class, put $V := \mathrm{Fix}(\tilde{\rho}^f) \cap S$, where $\tilde{\rho}$ is the Steinberg endomorphism of the ambient algebraic group with $\tilde{\rho}^2 = \varphi_2$ and $\tilde{\rho}^{2f} = F_q$. Thus $V = \mathrm{Sz}(q)$ for odd f , whereas for even f the group $V = \mathrm{Sp}(4, 2^{f/2})$ is the subfield subgroup over $\mathbb{F}_{2^{f/2}}$ corresponding to the degree-two extension $\mathbb{F}_{2^f}/\mathbb{F}_{2^{f/2}}$. In both cases V is maximal [1, Table 8.14, p. 384], [30].

Stability of $[V]$. Here $\mathrm{Aut}(S) = \mathrm{Inn}(S)\langle\rho\rangle$ with $\rho = \tilde{\rho}|_S$, and $\tilde{\rho}$ commutes with its own power $\tilde{\rho}^f$, so $\rho(V) = V$; inner automorphisms preserve $[V]$ trivially, so $[V]$ is Aut-stable.

Primes. Case f odd, $2f \neq 6$. Let r be a primitive prime divisor of $2^{2f} - 1$. Such a prime divides $q + 1$, so $v_r(|S|) = v_r((q^2 - 1)(q^4 - 1)) = 2v_r(q + 1) \geq 2$, while $v_r(|B|) = v_r(q^4(q - 1)^2) = 0$ and $v_r(|\mathrm{Sz}(q)|) = 0$ since $\gcd(q + 1, q(q^2 + 1)(q - 1)) = 1$ for q even; finally $x \mid 2f < r$. If $2f = 6$ (so $q = 8$), take $r = 3$; then $v_3(|\mathrm{Sp}(4, 8)|) = v_3(63 \cdot 4095) = 4 > 1 \geq v_3(x)$, with $v_3(|B|) = v_3(|\mathrm{Sz}(8)|) = 0$.

Case f even. Let r be a primitive prime divisor of $2^{4f} - 1$. Such a prime exists because $4f \geq 8$; it divides $q^2 + 1$, so $v_r(|S|) \geq 1$; $|\mathrm{Sp}(4, q_0)| = q_0^4(q_0^2 - 1)(q_0^4 - 1)$ with $q_0^2 = q$ has cyclotomic exponents $\leq 2f < 4f$, so $v_r(|V|) = 0$ and $v_r(|B|) = 0$; again $x \mid 2f < r$. The sweep-K3 certificate for $\mathrm{Sp}(4, 4)$ records the same construction: the pair (Borel, $\mathrm{Sp}(4, 2)$) at $r = 5 = \Phi_4(2)$ and $r = 17 = \Phi_8(2)$.

B6. $F_4(q)$, $q = 2^f$, X with graph part. Here ρ reverses the diagram $1 - 2 \Rightarrow 3 - 4$, swapping $1 \leftrightarrow 4$ and $2 \leftrightarrow 3$. The pair consists of the two novelties $([Q_{J_1}], [Q_{J_2}])$, where $J_1 := \{2, 3\}$ and $J_2 := \{1, 4\}$, both ρ -invariant node sets. The proper overgroups of Q_{J_1} are $Q_{\{1,2,3\}} = P_4$ and $Q_{\{2,3,4\}} = P_1$, swapped by ρ ; those of Q_{J_2} are $Q_{\{1,2,4\}}$ and $Q_{\{1,3,4\}}$, also swapped by ρ and of distinct types, hence in distinct classes. So Lemma 3.6 applies to both. The corresponding Levi types are B_2 and $A_1 \times A_1$. The first is isomorphic to $\mathrm{Sp}(4, q)$ and has exponents at most $4f$, while the second has exponents at most $2f$. Let r be a primitive prime divisor of $2^{12f} - 1$. It avoids both, and $v_r(|S|) \geq 1$ while $x \mid 2f < 12f < r$. There are no exceptions.

B7. $G_2(q)$, $q = 3^f$, X with graph part. Here ρ swaps the two nodes (long and short roots) and $\rho^2 = \varphi_3$. The pair again consists of $[B]$, a novelty whose overgroups P_1, P_2 are swapped by ρ (Lemma 3.6), and $[V]$, where $V := \mathrm{Fix}(\tilde{\rho}^f) \cap S$ as in B5. Thus $V = {}^2G_2(q)$ for f odd, maximal in $G_2(q)$ [19, Thm. A, p. 33] (for $f = 1$, $V = {}^2G_2(3) \cong \mathrm{P}\Gamma\mathrm{L}_2(8)$, maximal in $G_2(3)$ by the ATLAS [8] and [19, Thm. A, p. 33]), and $V = G_2(3^{f/2})$ for even f . In the latter case, V is the subfield subgroup over $\mathbb{F}_{3^{f/2}}$ corresponding to the degree-two extension $\mathbb{F}_{3^f}/\mathbb{F}_{3^{f/2}}$, and it too is maximal [19, Thm. A, p. 33]. The class $[V]$ is Aut-stable by the commuting-endomorphism argument of B5, since $\mathrm{Out}(G_2(3^f)) = \langle\rho\rangle \cong C_{2f}$.

Primes. Case f odd. Let r be a primitive prime divisor of $3^{3f} - 1$. Such a prime exists for all $f \geq 1$, because $p = 3 \neq 2$ excludes the 2^6 exception ($f = 1$ gives $r = 13 \mid 3^3 - 1$). Then $r \mid \Phi_3(q) = q^2 + q + 1$, which divides $q^6 - 1$. Since $|S| = q^6(q^6 - 1)(q^2 - 1)$, it follows that $r \mid |S|$. Primitivity gives $r \nmid |B| = q^6(q - 1)^2$ and $r \nmid q - 1$ (as $r \mid 3^f - 1$ would force $\mathrm{ord}_r(3) \leq f < 3f$), while $r \nmid |{}^2G_2(q)| = q^3(q^3 + 1)(q - 1)$ because $r \mid q^3 - 1$ and $\gcd(q^3 - 1, q^3 + 1) = 2 < r$. Since $x \mid 2f$ and $r \equiv 1 \pmod{3f}$, we have $r \geq 3f + 1 > 2f$; hence $v_r(x) = 0 < 1 \leq v_r(|S|)$.

Case f even. Let r be a primitive prime divisor of $3^{6f} - 1$. Such a prime divides $\Phi_6(q) = q^2 - q + 1$, which divides $q^3 + 1$; hence $r \nmid |B|$ and $r \nmid |G_2(3^{f/2})| = q^3(q^3 - 1)(q - 1)$, while $x \mid 2f < 6f < r$. ($G_2(3)$ additionally carries a full machine certificate for both X .)

In every case Theorem 4.1 excludes S^k for every $k \geq 2$. $\square$

7. PROOF OF THE MAIN THEOREM

Proposition 7.1 (Coverage). *Every non-abelian finite simple group is excluded in the sense of Convention 4.4.*

Proof. By the classification of finite simple groups [12, Ch. 1, Table I, pp. 8–10], S is alternating, of Lie type, or one of the 26 sporadic groups. The cases are as follows. The simplicity exceptions are those in [13, Thm. 2.2.7(a)]; the low-rank classical identifications and the exceptional derived-group identifications are recorded in [8, §2.4, pp. xi–xii, and §3.5, p. xv] (with the individual ATLAS entries pinpointed in the source map in [28]).

- A_n ($n \geq 5$): Theorem 6.2.
- $\text{PSL}(2, q)$ ($q \geq 4$): Theorem 6.1.
- $\text{PSL}(n, q)$ ($n \geq 3$), $D_n(q)$ ($n \geq 4$), $E_6(q)$: Theorem 6.6.
- $\text{PSp}(2n, q)$: $n \geq 3$, or $n = 2$ with q odd: Theorem 6.3; $\text{Sp}(4, 2^f)$, $f \geq 2$: Theorem 6.6 ($\text{Sp}(4, 2)$ is not simple).
- $\Omega(2n+1, q)$, q odd, $n \geq 3$: Theorem 6.3 (for q even, $B_n(q) \cong C_n(q)$, already listed; $\Omega(5, q) \cong \text{PSp}(4, q)$).
- $E_7(q)$, $E_8(q)$; $F_4(q)$, $p \neq 2$; $G_2(q)$, $p \neq 3$: Theorem 6.3. $F_4(2^f)$ and $G_2(3^f)$: Theorem 6.6. ($G_2(2)$ and ${}^2G_2(3)$ are not simple; their derived groups $\text{PSU}(3, 3)$ and $\text{PSL}(2, 8)$ are covered.)
- $\text{PSU}(n, q)$: $n = 3$: Theorem 6.5; $n \geq 4$: Theorem 6.4.
- ${}^2D_n(q)$ ($n \geq 4$), ${}^3D_4(q)$, ${}^2E_6(q)$, ${}^2F_4(2^f)$ (f odd ≥ 3): all four families are covered by Theorem 6.4; the Tits group ${}^2F_4(2)'$ by Proposition 5.2. (2D_2 , 2D_3 are $\text{PSL}(2, q^2)$, $\text{PSU}(4, q)$, already listed.)
- $\text{Sz}(2^f)$ (f odd ≥ 3), ${}^2G_2(3^f)$ (f odd ≥ 3): Theorem 6.5.
- Sporadic groups: Proposition 5.2.

Every finite simple group appears (repetitions from exceptional isomorphisms are harmless), so the proposition follows. $\square$

Proof of Theorem 1.1. Suppose not, and let G have least order among the non-soluble groups with property P. By Proposition 2.3, G has a unique minimal normal subgroup $N = S^k$, S non-abelian simple, and after conjugation $N \leq G \leq X \wr S_k$ as in Convention 2.4. If $k = 1$, then G is almost simple, contrary to the theorem of Tikhonenko–Tyutyaynov [32]. Hence $k \geq 2$. By Proposition 7.1 there are classes $[V] \neq [W]$, maximal in $\mathcal{P}_X(S)$ and normally saturating, and a prime p with $v_p(|S|) - v_p(|V|) - v_p(|W|) > v_p(x)$. By Theorem 4.1, G does not have property P, a contradiction. $\square$

Corollary 7.2 (Supplementary reproof). *The present criterion also proves that no almost simple group has property P.*

Proof. Apply Theorem 4.3 with the same witnesses used in Proposition 7.1. $\square$

Remark 7.3. The argument proves the following stronger statement: for *every* finite group G with a unique minimal normal subgroup S^k (S non-abelian simple, $k \geq 2$), some pair of non-conjugate maximal subgroups of G fails to factorize G . The reduction of §2 is needed only to pass from an arbitrary non-soluble group to this configuration.

Data Availability. All verification code, including the GAP scripts, and all certificate logs required to reproduce the machine-verified claims of §5, together with the arithmetic verification records for the family proofs of §6, are archived in the versioned release [28]. The release includes SHA-256 checksums of every certificate log, a CITATION.cff file, a single command that runs the quick verification suite, and separate documented commands for full GAP certificate regeneration and byte comparison, Lean and Rocq builds, and mutation-test reproduction. Its README maps each claim of this paper to the script and certificate that verify it. The priority-search record and source ledger supporting the discussion in the Introduction are also included.

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Declaration of generative AI and AI-assisted technologies in the research and writing process. During this project the author used Anthropic’s Claude (2026 Claude 5-family models) and OpenAI Codex for mathematical exploration, generation and critical evaluation of candidate arguments, formal-proof development, drafting and debugging the GAP and Python code, literature-search assistance, verification-workflow construction, and manuscript organization and editing. The author reviewed the mathematical arguments, reran the verification scripts or checked their outputs against the committed certificate logs, verified the cited references, and edited the final manuscript. All machine-verified claims are backed by the committed, independently checkable certificates described in §5. The author takes full responsibility for the content of this article.

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